Project Design Phase-I
Proposed Solution Template

Date	08 March 2026
Team ID	—
Project Name	Predictive Pulse – Hypertension Risk Detection using Machine Learning
Maximum Marks	2 Marks

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1	Problem Statement (Problem to be solved)	Hypertension is one of the most common cardiovascular conditions, yet many individuals are unaware of their risk until serious symptoms appear. Traditional diagnosis requires hospital visits and medical tests, which can be time-consuming. There is a lack of simple AI-based tools that allow individuals to quickly assess their hypertension risk based on symptoms and lifestyle factors.
2	Idea / Solution Description	Develop an AI-powered web application called **Predictive Pulse** that predicts hypertension risk using a **Machine Learning model (Logistic Regression)**. Users enter health information such as age group, symptoms, blood pressure levels, medication history, and lifestyle factors through a **Flask-based web interface**. The system processes this data, predicts the hypertension stage, and provides **risk level, confidence score, and medical recommendations** instantly.
3	Novelty / Uniqueness	Unlike traditional health websites that provide only general information, this system uses **Machine Learning for personalized prediction**. The web application integrates **AI prediction, symptom analysis, and medical recommendations in a single platform**. The interactive medical dashboard and real-time prediction capability make it more advanced than simple health calculators.
4	Social Impact / Customer Satisfaction	The system promotes **preventive healthcare and early detection of hypertension**, helping users take action before serious health complications occur. It can benefit **patients, healthcare students, and health-conscious individuals** by providing quick insights about their blood pressure risk and encouraging healthier lifestyle choices.
5	Business Model (Revenue Model)	The platform can operate as a **freemium digital health tool**. Basic hypertension risk prediction can remain free for users, while advanced features such as **detailed health reports, historical tracking, and doctor consultation integration** can be offered through subscription. Hospitals and wellness platforms can integrate the system as a **health monitoring tool** for patients.
6	Scalability of the Solution	The machine learning model can be improved with larger medical datasets and additional features. The Flask application can be deployed on cloud platforms to support large numbers of users. Future versions may include **mobile applications, wearable device integration, and advanced AI models** for more accurate health predictions.